EEE4114F: Digital Signal Processing

Class Test

15 March 2022

SOLUTIONS

Please do not put your name or student number anywhere on this script.

Peoplesoft ID:

Information

- The test is closed-book.
- This test has *four* questions, totalling 20 marks.
- Answer *all* the questions.
- You have 60 minutes.
- An information sheet is attached.

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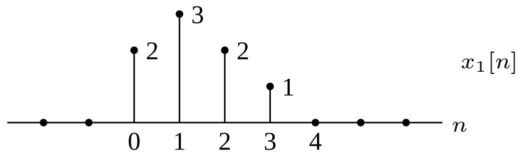
1. (5 marks) A discrete-time system is described by the following rule:

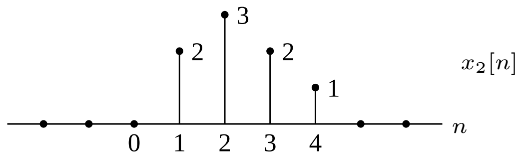
$$y[n] = 0.5x[2n] + 0.5x[2n-1],$$

where $x[n]$ is the input and $y[n]$ is the output.

(a) Sketch the output $y_1[n]$ when the input is $x_1[n]$ below:



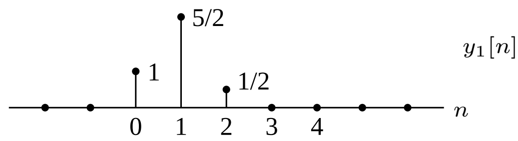
(b) Sketch the output $y_2[n]$ when the input is $x_2[n]$ below:



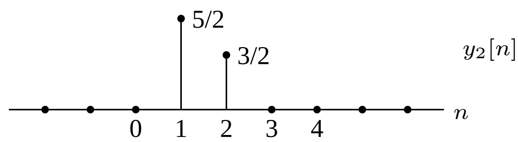
(c) Is the system causal?

(d) Is the system time invariant?

(a) Output as follows:



(b) Output as follows:



(c) Since $y[1] = 0.5x[2] + 0.5x[1]$ we see that we need to know $x[2]$ to calculate $y[1]$. Thus the system is not causal.

(d) We observe that $x_2[n] = x_1[n-1]$ but $y_2[n] \neq y_1[n-1]$. Thus the system is not time invariant.

2. (5 marks) Suppose that

$$y[n] = u[n-1] * 2^n u[n].$$

Plot $y[n]$ over the range $-5 \leq n \leq 5$.

We can find an expression for $y[n]$ although there are other ways to proceed. The required convolution is

$$y[n] = \sum_{k=-\infty}^{\infty} 2^k u[k] u[n-k-1] = \sum_{k=0}^{\infty} 2^k u[(n-1)-k].$$

For $n-1 < 0$ (or $n < 1$) this is zero since all the terms in the sum are zero. For $n \geq 1$ we have the sum of a finite geometric progression

$$y[n] = \sum_{k=0}^{n-1} 2^k = \frac{1-2^n}{1-2} = 2^n - 1.$$

Over the specified range the only nonzero values are $y[1] = 1$, $y[2] = 3$, $y[3] = 7$, $y[4] = 15$, and $y[5] = 31$.

3. (5 marks) Consider a system with the following impulse response:

$$h[n] = (1/3)^n u[n] + (1/2)^{n-2} u[n-1].$$

(a) Determine the transfer function $H(e^{j\omega})$ of the system.

(b) Find a difference equation representation of the system.

(a) Since $(1/2)^{n-2}u[n-1] = (1/2)^{-1}(1/2)^{n-1}u[n-1]$ we can find the z-transform:

$$H(z) = \frac{1}{1 - 1/3z^{-1}} + (1/2)^{-1}\frac{z^{-1}}{1 - 1/2z^{-1}} = \frac{1 + 3/2z^{-1} - 2/3z^{-2}}{(1 - 1/3z^{-1})(1 - 1/2z^{-1})}$$

with ROC $|z| > 1/2$. The system function converges on the unit circle so the transfer function is

$$H(e^{j\omega}) = \frac{1 + 3/2e^{-j\omega} - 2/3e^{-j2\omega}}{(1 - 1/3e^{-j\omega})(1 - 1/2e^{-j\omega})}.$$

(b) Since

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1 + 3/2z^{-1} - 2/3z^{-2}}{1 - 5/6z^{-1} + 1/6z^{-2}}$$

we have $(1 - 5/6z^{-1} + 1/6z^{-2})Y(z) = (1 + 3/2z^{-1} - 2/3z^{-2})X(z)$ and therefore

$$y[n] - 5/6y[n-1] + 1/6y[n-2] = x[n] + 3/2x[n-1] - 2/3x[n-2].$$

4. (5 marks) Suppose a stable and causal system has the following z-transform:

$$H(z) = \frac{1 + 0.8z^{-1}}{1 + 0.4z^{-1} - 0.12z^{-2}}.$$

(a) Determine the first 4 samples of the impulse response of $H(z)$.

(b) Does a stable and causal inverse system $G(z)$ exist? If so, specify $G(z)$.

(a) Since $Y(z)(1 + 0.4z^{-1} - 0.12z^{-2}) = X(z)(1 + 0.8z^{-1})$ the following difference equation arises:

$$y[n] = -0.4y[n-1] + 0.12y[n-2] + x[n] + 0.8x[n-1].$$

The impulse response $h[n]$ (zero for $n < 0$ since causal) is the output when the input is $x[n] = \delta[n]$, so

$$h[n] = -0.4h[n-1] + 0.12h[n-2] + \delta[n] + 0.8\delta[n-1].$$

Thus

$$\begin{aligned} h[0] &= 1 \\ h[1] &= -0.4(1) + 0.8 = 0.4 \\ h[2] &= -0.4(0.4) + 0.12 = -0.04 \\ h[3] &= -0.4(-0.04) + 0.12(0.4) = 0.0640. \end{aligned}$$

One could also do a partial fraction expansion to get

$$H(z) = -1/4\frac{1}{1 + 0.6z^{-1}} + 5/4\frac{1}{1 - 0.2z^{-1}}.$$

Thus $h[n] = -1/4(-0.6)^n u[n] + 5/4(0.2)^n u[n]$ and the required values can be calculated.

(b) The system function factorises as

$$H(z) = \frac{1 + 0.8z^{-1}}{(1 + 0.6z^{-1})(1 - 0.2z^{-1})} = \frac{z(z + 0.8)}{(z + 0.6)(z - 0.2)}$$

which has poles at $z = -0.6$ and $z = 0.2$, and zeros at $z = 0$ and $z = -0.8$. Since the poles and zeros are all inside the unit circle the system has a causal and stable inverse of

$$G(z) = \frac{1 + 0.4z^{-1} - 0.12z^{-2}}{1 + 0.8z^{-1}}$$

with ROC $|z| > 0.8$.

Transforms

$$X(j\Omega) = \int_{-\infty}^{\infty} x(t)e^{-j\Omega t}dt \quad \Longleftrightarrow \quad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\Omega)e^{j\Omega t}d\Omega \qquad \text{(CTFT)}$$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\omega})e^{j\omega n}d\omega \qquad \text{(DTFT)}$$

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n} \quad \Longleftrightarrow \quad x[n] = \frac{1}{2\pi j}\oint_C X(z)z^{n-1}dz \qquad \text{(ZT)}$$

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn} \quad \Longleftrightarrow \quad x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]W_N^{-kn} \quad \text{with} \quad W_N = e^{-j\frac{2\pi}{N}} \qquad \text{(DFT)}$$

Discrete-time Fourier transform properties

Sequences $x[n]$, $y[n]$	Transforms $X(e^{j\omega})$, $Y(e^{j\omega})$	Property
$ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$	Linearity
$x[n - n_d]$	$e^{-j\omega n_d}X(e^{j\omega})$	Time shift
$e^{j\omega_0 n}x[n]$	$X(e^{j(\omega-\omega_0)})$	Frequency shift
$x[-n]$	$X(e^{-j\omega})$	Time reversal
$nx[n]$	$j\frac{dX(e^{j\omega})}{d\omega}$	Frequency diff.
$x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$	Convolution
$x[n]y[n]$	$\frac{1}{2\pi}\int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega-\theta)})d\theta$	Modulation

Common discrete-time Fourier transform pairs

Sequence	Fourier transform
$\delta[n]$	1
$\delta[n - n_0]$	$e^{-j\omega n_0}$
$1 \quad (-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega + 2\pi k)$
$a^n u[n] \quad (\lvert a\rvert < 1)$	$\frac{1}{1-ae^{-j\omega}}$
$u[n]$	$\frac{1}{1-e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega + 2\pi k)$
$(n+1)a^n u[n] \quad (\lvert a\rvert < 1)$	$\frac{1}{(1-ae^{-j\omega})^2}$
$\frac{\sin(\omega_c n)}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1 & \lvert\omega\rvert < \omega_c \\ 0 & \omega_c < \lvert\omega\rvert \le \pi \end{cases}$
$x[n] = \begin{cases} 1 & 0 \le n \le M \\ 0 & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)}e^{-j\omega M/2}$
$e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 + 2\pi k)$

Z-transform properties

Sequences $x[n]$, $y[n]$	Transforms $X(z)$, $Y(z)$	ROC	Property
$ax[n] + by[n]$	$aX(z) + bY(z)$	ROC contains $R_x \cap R_y$	Linearity
$x[n - n_d]$	$z^{-n_d} X(z)$	ROC $= R_x$	Time shift
$z_0^n x[n]$	$X(z/z_0)$	ROC $= \|z_0\|R_x$	Frequency scale
$x^*[-n]$	$X^*(1/z^*)$	ROC $= \frac{1}{R_x}$	Time reversal
$nx[n]$	$-z\frac{dX(z)}{dz}$	ROC $= R_x$	Frequency diff.
$x[n] * y[n]$	$X(z)Y(z)$	ROC contains $R_x \cap R_y$	Convolution
$x^*[n]$	$X^*(z^*)$	ROC $= R_x$	Conjugation

Common z-transform pairs

Sequence	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{1}{1-z^{-1}}$	$\|z\| > 1$
$-u[-n-1]$	$\frac{1}{1-z^{-1}}$	$\|z\| < 1$
$\delta[n-m]$	z^{-m}	All z except 0 or ∞
$a^n u[n]$	$\frac{1}{1-az^{-1}}$	$\|z\| > \|a\|$
$-a^n u[-n-1]$	$\frac{1}{1-az^{-1}}$	$\|z\| < \|a\|$
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| > \|a\|$
$-na^n u[-n-1]$	$\frac{az^{-1}}{(1-az^{-1})^2}$	$\|z\| < \|a\|$
$\begin{cases} a^n & 0 \le n \le N-1, \\ 0 & \text{otherwise} \end{cases}$	$\frac{1-a^N z^{-N}}{1-az^{-1}}$	$\|z\| > 0$
$\cos(\omega_0 n)u[n]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}}$	$\|z\| > 1$
$r^n \cos(\omega_0 n)u[n]$	$\frac{1-r\cos(\omega_0)z^{-1}}{1-2r\cos(\omega_0)z^{-1}+r^2z^{-2}}$	$\|z\| > \|r\|$

Other

$$\sum_{n=0}^{N-1} a^n = \frac{1 - a^N}{1 - a} \quad \text{for} \quad a \neq 1.$$